LAB EXPERIMENT RECORD

Set 17: Employee Performance Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	10/10/2026
Day	Saturday

AIM

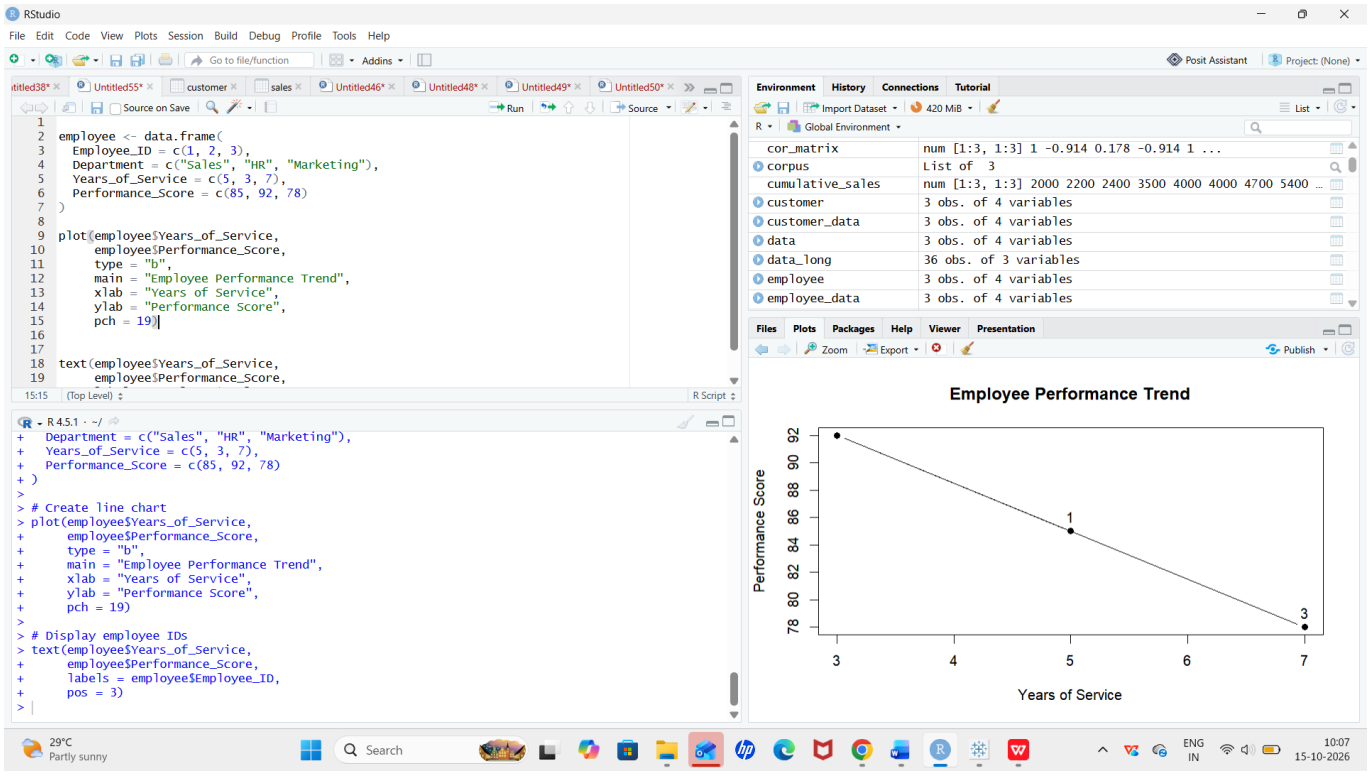
To analyze employee performance data using Tableau by creating a line chart to visualize performance scores based on years of service, a bar chart to show employee distribution across departments, a table to display employee details, and an interactive dashboard combining all visualizations.

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

plot(employee$Years_of_Service,
     employee$Performance_Score,
     type = "b",
     main = "Employee Performance Trend",
     xlab = "Years of Service",
     ylab = "Performance Score",
     pch = 19)


text(employee$Years_of_Service,
     employee$Performance_Score,
     labels = employee$Employee_ID,
     pos = 3)
```



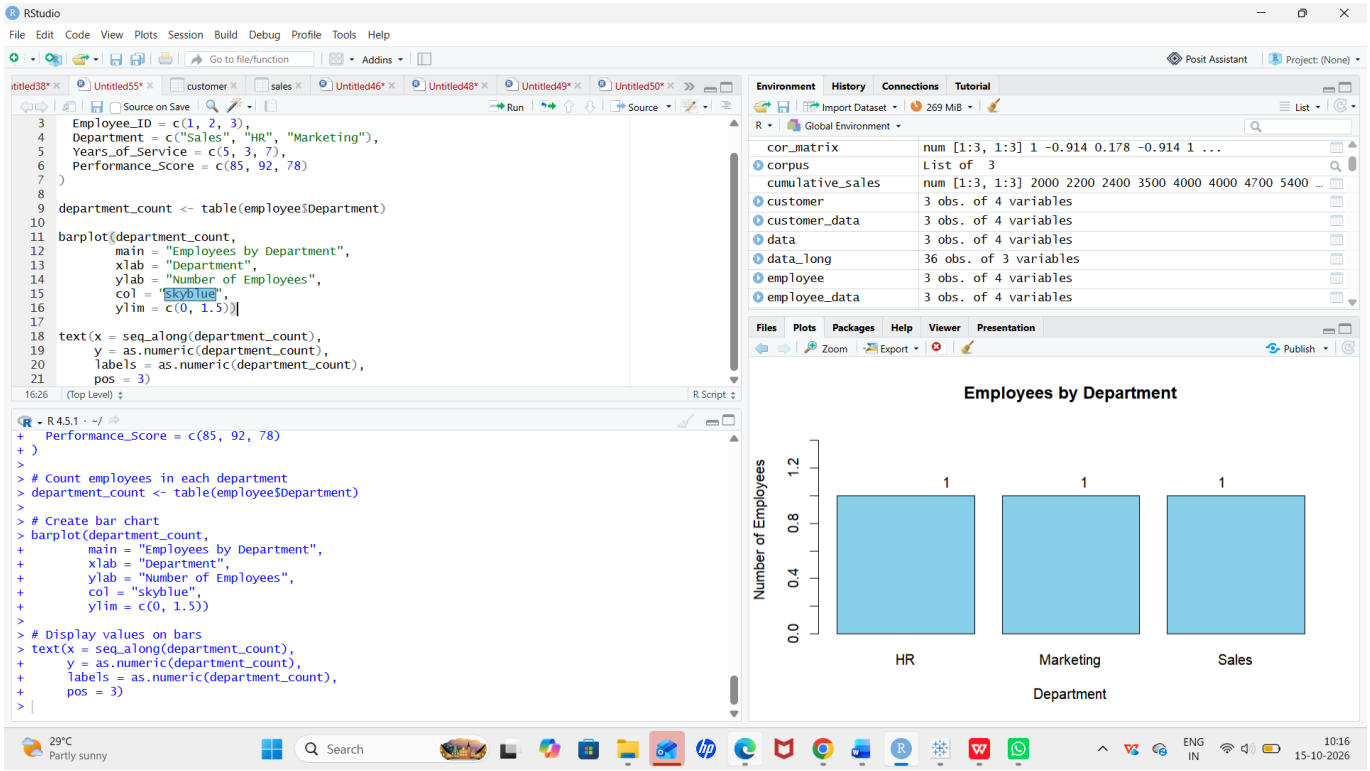
2. Generate a bar chart showing the distribution of employees across different departments.

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

department_count <- table(employee$Department)

barplot(department_count,
        main = "Employees by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = "skyblue",
        ylim = c(0, 1.5))

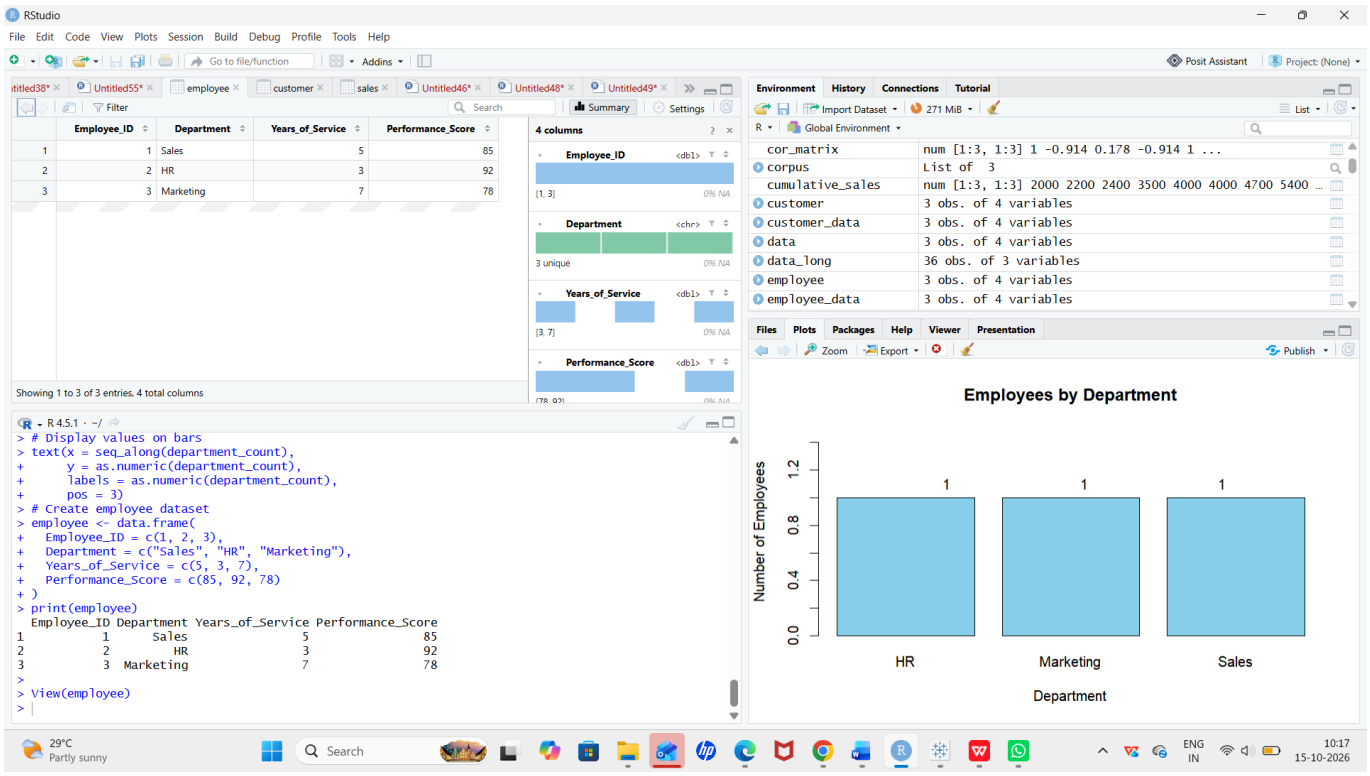
text(x = seq_along(department_count),
     y = as.numeric(department_count),
     labels = as.numeric(department_count),
     pos = 3)
```



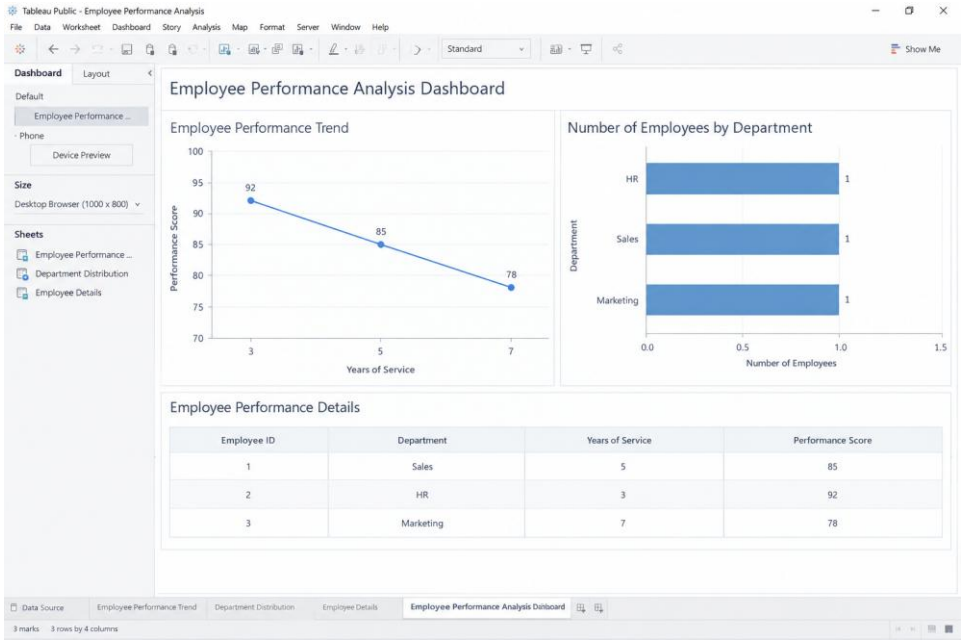
3. Build a table to display the performance data for each employee. Label the table elements.

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
print(employee)

View(employee)
```



4.Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive



Procedure

1. Prepare the employee performance dataset with Employee ID, Department, Years of Service, and Performance Score.

2. Open Tableau and connect the employee dataset using the Text File option.

3. Create a line chart by placing Years of Service on Columns and Performance Score on Rows. Add axis labels and a suitable chart title.

4. Create a bar chart by placing Department on Columns and the count of Employee ID on Rows. Display mark labels and add a suitable title.

5. Create a text table displaying Employee ID, Department, Years of Service, and Performance Score.

6. Create a new dashboard and combine the line chart, bar chart, and employee details table.

7. Adjust the layout, add a dashboard title, and save the Tableau workbook.

Result

Thus, the employee performance data was successfully visualized using Tableau. The line chart displayed performance scores according to years of service, the bar chart showed employee distribution across departments, and the table presented employee details. An interactive dashboard was created by combining all three visualizations for effective employee performance analysis.